Product Requirements Document (PRD)

CareQueue - Hospital Digital Token and Queue Management System

Field	Value
Document	Product Requirements Document
Product	CareQueue - Hospital Digital Token and Queue Management System
Problem statement	Patients wait in queues for registration and billing. Build a digital token and queue-management system that displays the current token and estimated waiting time.
Version / Status	v1.0 - Draft for hackathon submission
Date	24 October 2026
Team	Future Coders
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1. Goal

Deliver a working MVP where a receptionist issues tokens, a waiting-room screen shows the **current token** and each patient's **estimated waiting time**, and the receptionist calls the next patient with one click. Users and needs are defined in the MRD.

2. Features Ranked (MoSCoW)

2.1 Must Have (3)

#	Feature	Acceptance criteria
M1	**Token generation** - receptionist enters name and picks queue (Registration R / Billing B); system issues sequential token (R-001, B-001), resetting daily	Token created in one form submit; numbers never repeat within a day; slip/screen shows token + position
M2	**Call Next + live Now-Serving display** - one button advances the queue; a full-screen display shows the current token per queue and the next 5 waiting	Display updates without refresh within 2 s of Call Next; shows token numbers only (no names)
M3	**Estimated waiting time (EWT)** - shown for every waiting token on the display	EWT recalculates on every queue event; uses real service times after 3 tokens are served

EWT logic (M3): EWT = R + N x AST, where **AST** = average service time of the last 10 served tokens in that queue (default 3 min until 3 are served), **N** = tokens ahead, **R** = time left for the patient being served = max(0, AST - time at counter). Example: AST 4 min, 5 ahead, R 3 min -> **23 min**.

2.2 Should Have

- **S1** Skip / Recall for absent patients.
- **S2** Voice + chime announcement of called token (browser text-to-speech).
- **S3** QR code on token to track position and EWT on own phone (no app).
- **S4** "Send to Billing" - moves a registered patient into the billing queue with a linked B-token.

- **S5** Multiple counters per queue (EWT divides by open counters).

2.3 Could Have

- **C1** Priority tokens for seniors / emergencies.
- **C2** Daily stats: tokens served, average wait, busiest hour.
- **C3** Display in a local language (Tamil / Hindi).
- **C4** SMS / WhatsApp alert when 2 patients are ahead.

2.4 Won't Have (this version)

- Online / remote queue booking, doctor appointment scheduling.
- Payments or billing calculations.
- Patient medical records or HIS integration.
- Multi-branch / multi-clinic management.

3. Success Metrics

Metric	Target	How measured
Token issue time	Median 15 s or less per patient	Timestamp from form open to token created (app log)
Display update latency	95% of updates within 2 s	Compare Call Next server time vs display render time
EWT accuracy	80% of tokens within +/- 5 min of actual wait	Actual wait (created -> called) vs EWT shown at creation, after first 3 tokens
Counter crowding	50% fewer people standing at the counter at peak	Headcount at the same hour, before vs with CareQueue (pilot)
"How long?" questions	50% fewer per hour	Receptionist tally sheet, 1 hour before vs with CareQueue
Staff learning time	New receptionist issues and calls tokens alone within 5 min	Timed usability test, no training

4. Out of Scope

Excluded	Reason
Doctor consultation queues	Problem statement covers registration and billing only
Payments / bill amounts	Separate system; adds compliance burden
Medical records, ABHA / HIS integration	Privacy and integration effort beyond hackathon
Native mobile apps	Web page via QR is enough; no install friction
Remote / online token booking	Changes the walk-in model; future roadmap
Multi-branch, multi-language beyond one local language	Not needed for a single small clinic

5. Dependencies & Notes

- Suggested stack: React or HTML/JS, Node.js + Socket.IO (or Firebase) for real-time updates, Web Speech API for voice, qrcode.js for QR.
- Display must never show patient names (privacy).
- Assumptions behind these requirements are listed in MRD Section 6.